MIL-M-28787/48A(EC)
12 February 1979
SUPERSEDING
MIL-M-28787/48(NAVY)
8 November 1973

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, TRANSLATE, DIGITAL, KEY CODE MDP

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic translate driver module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Driver, translate
Family	Digital
Key code	MDP
Size	1A
Weight	35 grams maximum
Failure rate	3.1 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage	+5.0 Vdc ±10 percent -5.0 Vdc ±10 percent
Maximum power dissipation	
All outputs at low level	0.71 W
All outputs at high level	0.66 W
Input threshold voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	1.73 mA maximum
High level (I_{IH} = 1.5 mA)	40 µA maximum

Beneficaial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output voltage	
Low level (I_{IL} = 0.5 mA)	-3 Vdc minimum, -5.0 Vdc maximum
High level (I_I = 1.5 mA)	0 Vdc minimum, -0.5 Vdc maximum
Propagation delay time	
t_{PHL}	160 ns maximum
t_{PLH}	160 ns maximum

1.3 Module function. The translate driver module, designated as key code MDP, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains four negative interface line driver circuits. It is designed to drive up to a maximum of 150 ft of 2U cable (26 AWG twisted shielded pair) while also translating from positive to negative voltage levels. When used with twisted shielded pair, the signal side of the line should be connected to the module output, the low side of the line to signal ground, and the shield to chassis. The MDP module is intended for use with translate receiver module MDQ or its equivalent. One MDP may drive one MDQ. The combination is noninverting (from positive voltage levels of MDP inputs to positive voltage levels of MDQ output). Each driver circuit will withstand, without failure, a short between the driver output and ground, With power to the module removed, the driver output will present a minimum impedance of 100 kΩ to a negative voltage on the line between 0 and -7 V.

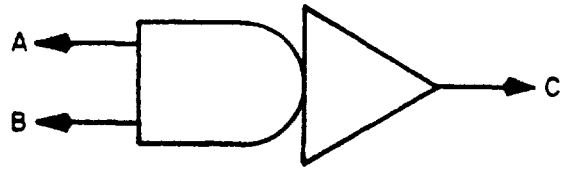
1.3.1 Input to output relationship. Each translate driver has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

Inputs		Output
A	B	C
L	L	L
L	H	L
H	L	L
H	H	H

NOTES:

1. Logic function: C = AB.
2. L = Low-voltage level.
3. H = High-voltage level.



1.3.2 Parallel outputs. Paralleling of outputs is not permissible.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pa 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.1 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MDP.

3.5.2 Size. The module shall be be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin of increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

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3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/48, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier.)

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	5
Pulse generator	Item J01	1
DC ammeter	Item X01	2
Megohm bridge	Item R06	1
Capacitance meter	Item C05	1
DC voltmeter	Item V03	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Square wave generator	Item H01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y02	1
Resistors		
51Ω ±5%, 1 W		2
147Ω ±1%, 1/4 W		4
5.9 kΩ ±1%, 1/4 W		4
2.05 kΩ ±1%, 1/4 W		1
3.16 kΩ ±1%, 1/4 W		1
51Ω ±1%, 1/4 W		1
3.01 kΩ ±1%, 1/4 W		1
5.11 kΩ ±1%, 1/4 W		1
511Ω ±1%, 1/4 W		1
1.87 kΩ ±1%, 1/4 W		1

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Equipment	No. required (maximum per test figure)
Capacitors	
0.01 µF ±10%	4
0.1 µF ±10%	1
Value as required for figure 18, note 4	1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through M, S, and T, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

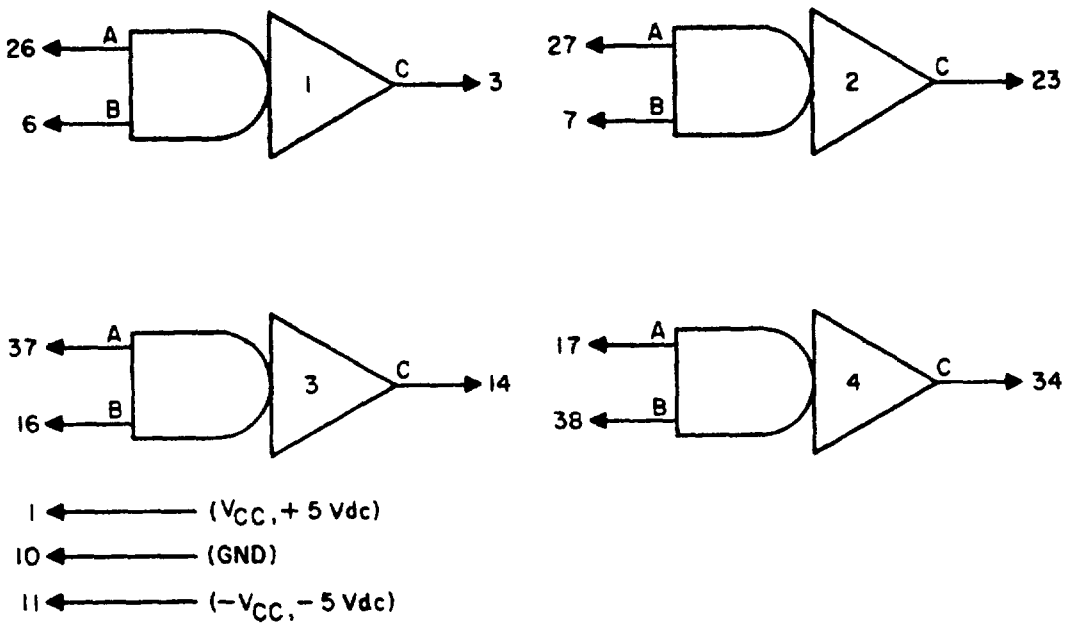
4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test condition S, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through M, S and T, shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements table II. Following the return of the module to 25°C, electrical tests B through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE: Unused pins: 2, 4, 5, 8, 9, 12, 13, 15, 18, 19, 20, 21, 22, 24, 25, 28, 29, 30, 31, 32, 33, 35, 36, 39, and 40.

FIGURE 1. MDP module function diagram, translate driver.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	Input breakdown current	III	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
B1	Isolation	IV	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B2-B26	Isolation, unused pins	IV	Q	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
C1-C8	Input capacitance	V	Q	2	15	2	15	2	15	2	15	pF
D1-D4	Output voltage, high level	VI	X	0.0	-0.5	0.0	-0.5	0.0	-0.5	0.0	-0.5	V
E1-E8	Output voltage, low level	VII	X	-3.0	-5.0	-3.0	-5.0	-3.0	-5.0	-3.0	-5.0	V
F1-F8	Input current, low level	VIII	X	0.4	1.73	0.4	1.73	0.4	1.73	0.4	1.73	mA
G1-G8	Input current, high level	IX	X	0	30	0	30	0	40	0	40	µA
H1	Power supply current, high level, $+V_{CC}$	X	S	1	92	1	92	1	92	1	92	mA
H2	Power supply current, high level, $-V_{CC}$	X	S	1	28	1	28	1	28	1	28	mA
H3	Power supply current, low level, $+V_{CC}$	X	S	1	80	1	80	1	80	1	80	mA
H4	Power supply current, low level, $-V_{CC}$	X	S	1	48	1	48	1	48	1	48	mA
J1-J4	Transition time (t_{TLH})	XI	S	4	40	4	40	4	50	4	50	ns
K1-K4	Transition time (t_{THL})	XI	S	4	40	4	40	4	50	4	50	ns
L1-L4	Propagation delay time (t_{PHL})	XI	S	25	150	25	150	25	160	25	160	ns
M1-M4	Propagation delay time (t_{PLH})	XI	S	25	150	25	150	25	160	25	160	ns
N1	Power supply transient amplitude, $+V_{CC}$	XII	Q	0	600	0	600	0	600	0	600	mV
N2	Power supply transient amplitude, $-V_{CC}$	XII	Q	0	500	0	500	0	500	0	500	mV

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1-P4	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	mV
R1-R4	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	mV
S1, S2	Power supply transient rise time	XV	S	8	12	8	12	8	12	8	12	ms
T1-T4	Output impedance check	XVI	Q	0	70	0	70	0	70	0	70	µA

NOTES:

1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

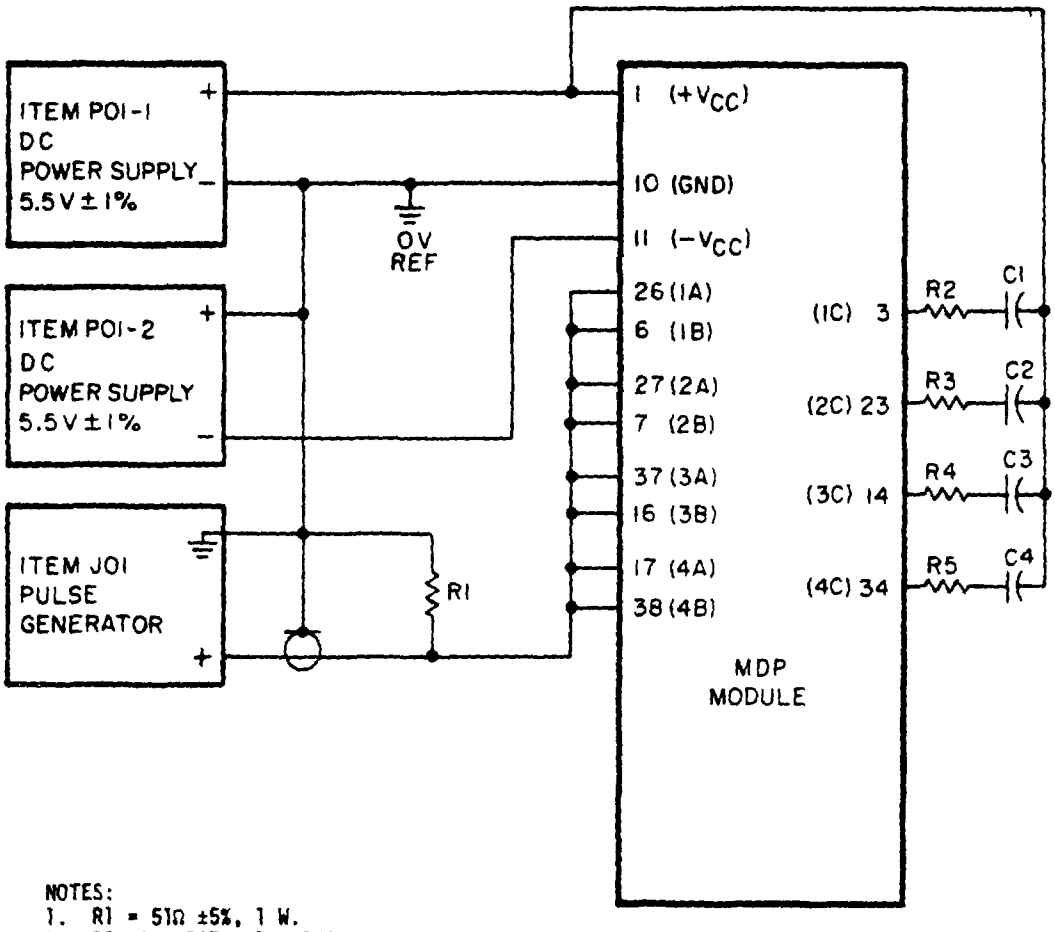
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N021)

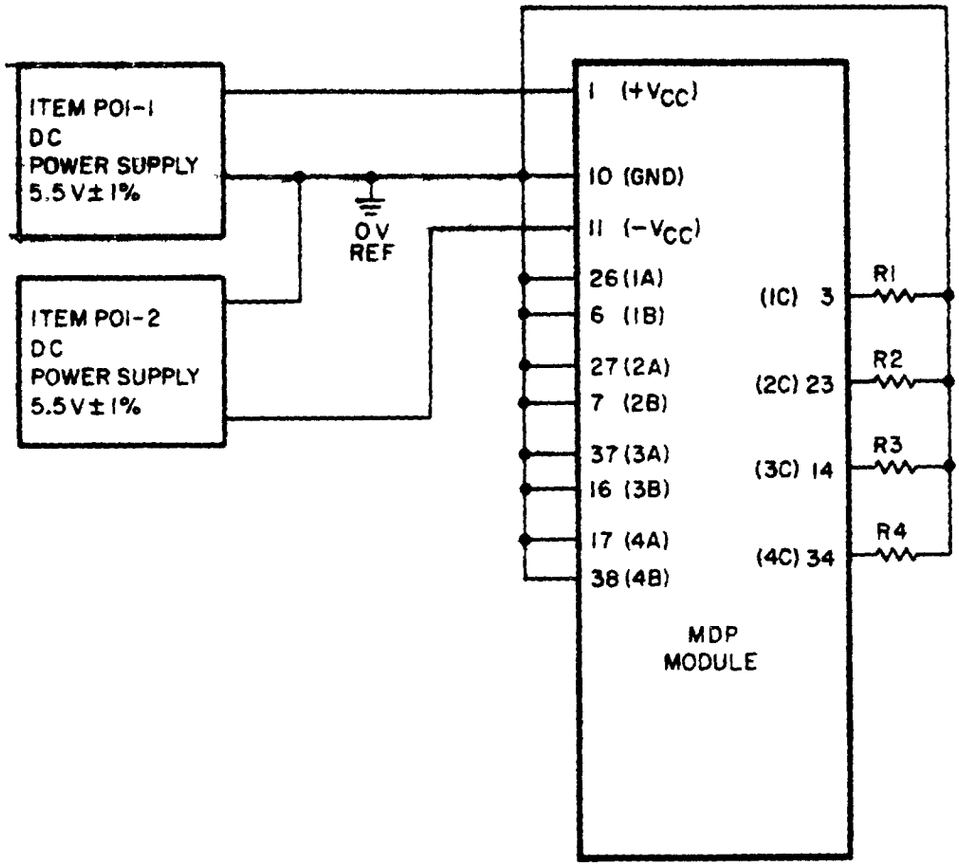
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2-R5 = 147Ω ±1%, 1/4 W.
3. C1-C4 = 0.01 µF ±10%.
4. Item JOI input pulse parameters:
   PA = 3.0 ±0.1 V.
   PW = 1 µs ±10%.
   PRR = 0.5 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 25 ns ±10%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R4 = 5.9 kΩ ±1%, 1/4 W.

FIGURE 3. Transient thermal test circuit.

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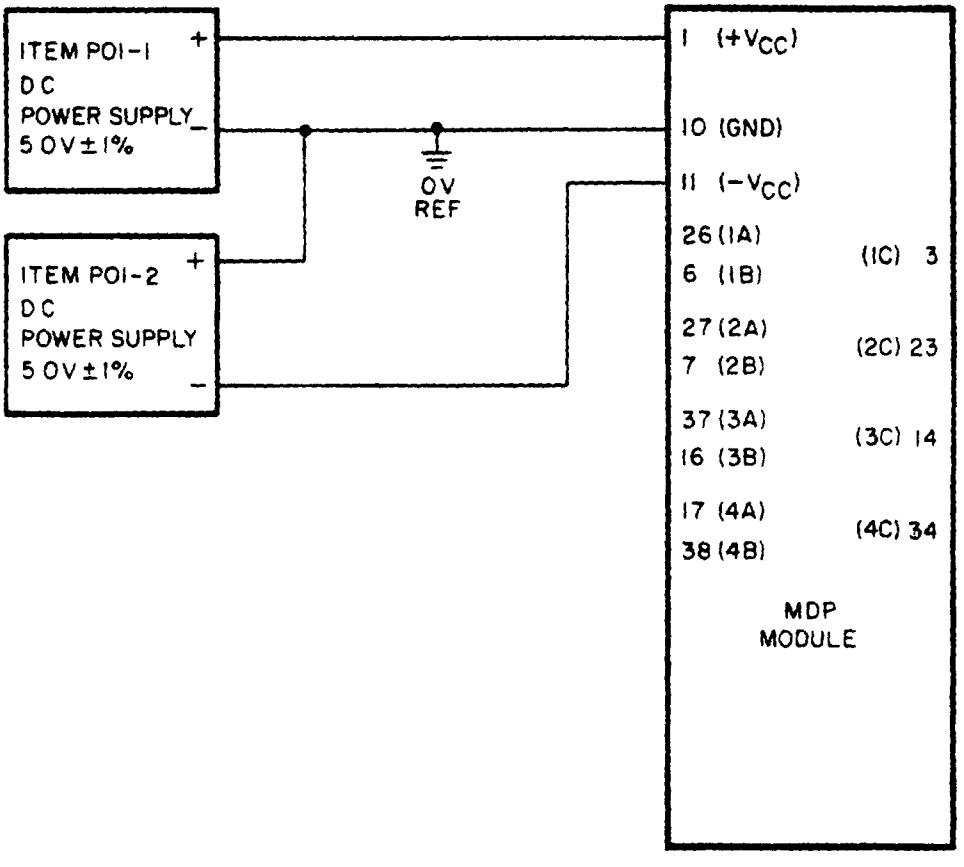


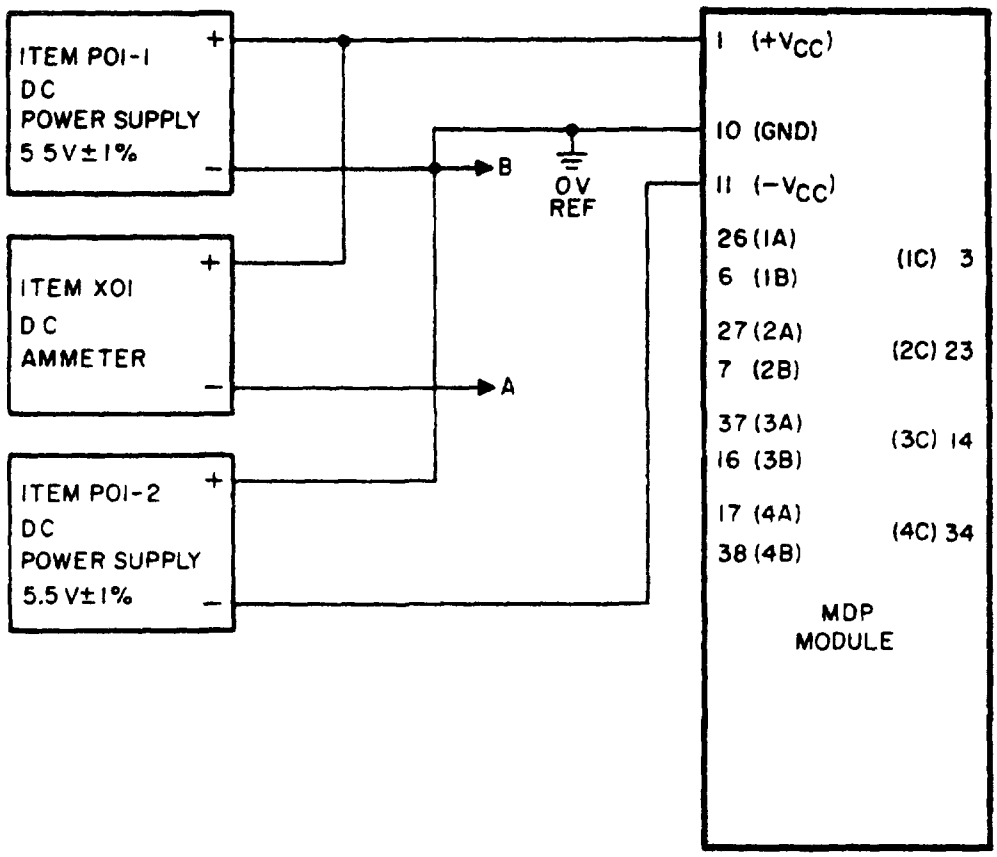
FIGURE 4. Vibration test circuit.

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TABLE III. Input breakdown current test.

Test condition		Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
A1	1A	Input breakdown current	26	6	0.0	1.0	mA	5
A2	1B		6	26				
A3	2A		27	7				
A4	2B		7	27				
A5	3A		37	16				
A6	3B		16	37				
A7	4A		17	38				
A8	4B	Input breakdown current	38	17	0.0	1.0	mA	5

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

FIGURE 5. Input breakdown current test circuit for table III.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation	C, 2,4,5,8,9, 12,13,15,18,19, 20,21,22,24,25, 28,29,30,31,32, 33,35,36,39,40	–	10	∞	MΩ	6
B2		C	2,4,5,8,9,12, 13,15,18,19, 20,21,22,24, 25,28,29,30, 31,32,33,35, 36,39,40				
B3		2	4,5,8,9,12 13,15,18,19, 20,21,22,24, 25,28,29,30, 31,32,33,35, 36,39,40				
B4		4	5,8,9,12,13, 15,18,19,20, 21,22,24,25, 28,29,30,31, 32,33,35,36, 39,40				
B5	Isolation	5	8,9,12,13,15, 18,19,20,21, 22,24,25,28, 29,30,31,32, 33,35,36,39, 40	10	∞	MΩ	6

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TABLE IV. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B6	Isolation	8	9,12,13,15, 18,19,20,21, 22,24,25,28, 29,30,31,32, 33,35,36,39, 40	10	-	MΩ	6
B7		9	12,13,15,18, 19,20,21,22, 24,25,28,29, 30,31,32,33, 35,36,39,40				
B8		12	13,15,18,19, 20,21,22,24, 25,28,29,30, 31,32,33,35, 36,39,40				
B9		13	15,18,19,20, 21,22,24,25, 28,29,30,31, 32,33,35,36, 39,40				
B10	Isolation	15	18,19,20,21, 22,24,25,28, 29,30,31,32, 33,35,36,39, 40	10	-	MΩ	6

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TABLE IV. Isolation test - Continued.

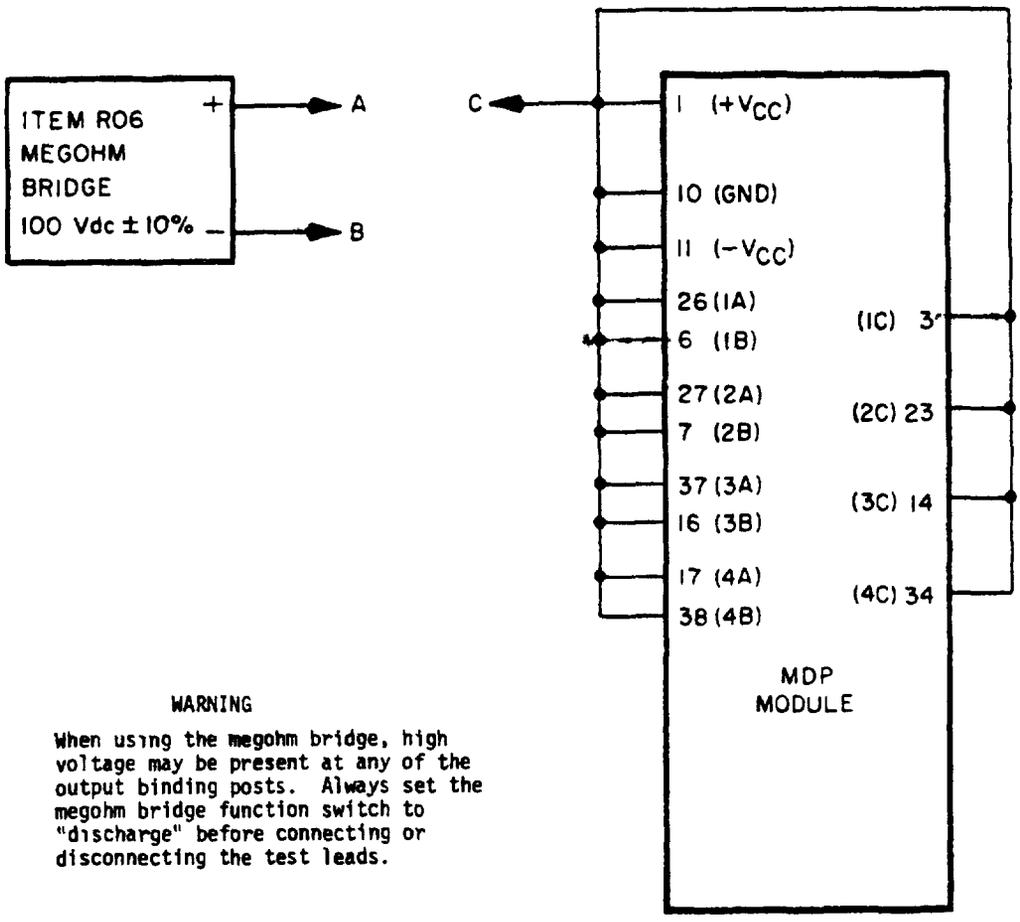
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B11	Isolation	18	19,20,21,22, 24,25,28,29, 30,31,32,33, 35,36,39,40	10	-	MΩ	6
B12		19	20,21,22,24, 25,28,29,30, 31,32,33,35, 36,39,40				
B13		20	21,22,24,25, 28,29,30,31, 32,33,35,36, 39,40				
B14		21	22,24,25,28, 29,30,31,32, 33,35,36,39, 40				
B15		22	24,25,28,29, 30,31,32,33, 35,36,39,40				
B16		24	25,28,29,30, 31,32,33,35, 36,39,40				
B17	Isolation	25	28,29,30,31, 32,33,35,36, 39,40	10	-	MΩ	6

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TABLE IV. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B18	Isolation	28	29,30,31,32, 33,35,36,39, 40	10	-	MΩ	6
B19		29	30,31,32,33, 35,36,39,40				
B20		30	31,32,33,35, 36,39,40				
B21		31	32,33,35,36, 39,40				
B22		32	33,35,36,39, 40				
B23		33	35,36,39,40				
B24		35	36,39,40				
B25		36	39,40				
B26	Isolation	39	40	10	-	MΩ	6

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE For test condition B1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

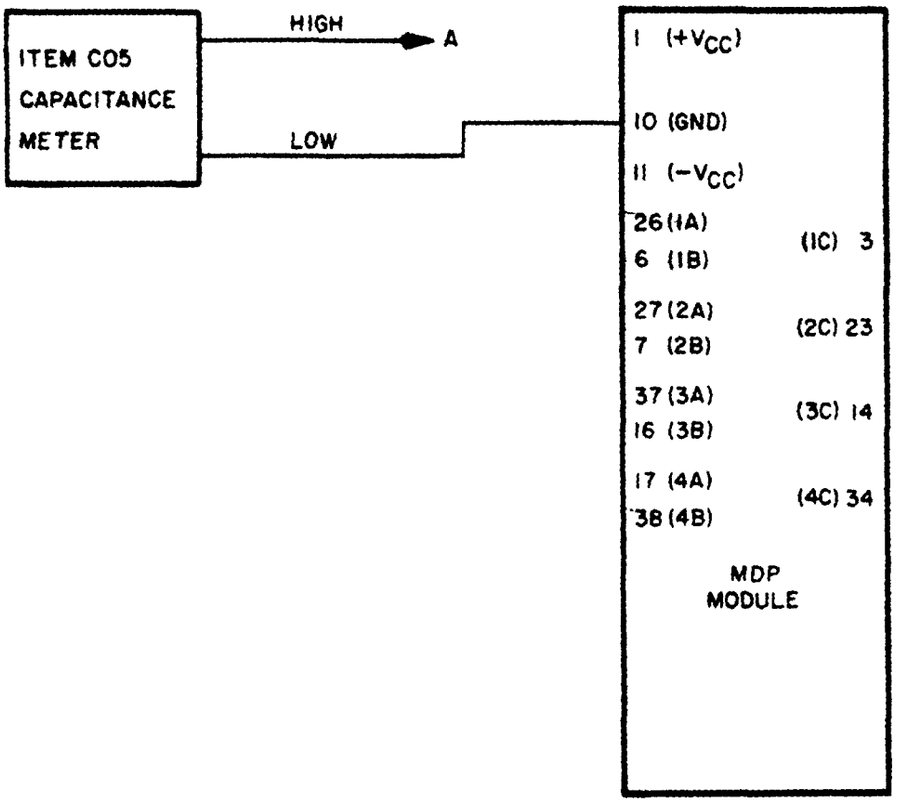
FIGURE 6. Isolation test circuit for table IV.

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TABLE V. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
C1	Input capacitance	26	2	15	pF	7
C2		6				
C3		27				
C4		7				
C5		37				
C6		16				
C7		17				
C8	Input capacitance	38	2	15	pF	7

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NOTES:
1. Null out the effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 7. Input capacitance test circuit for table V.

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TABLE VI. Output voltage (high-level) test.

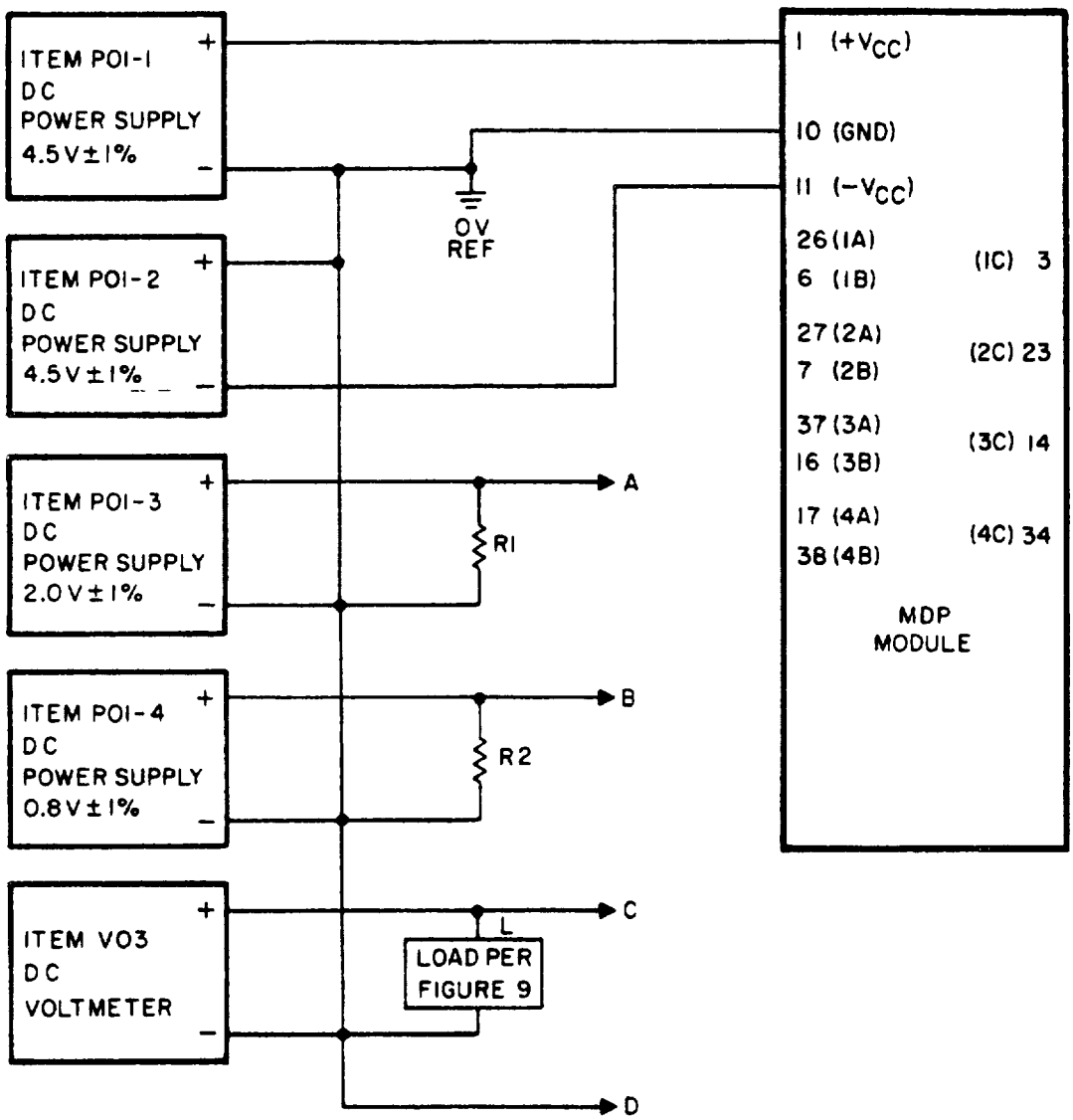
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	1C	Output voltage, high level	26,6	Open	3	0.0	-0.5	V	8
D2	2C	↓	27,7	↓	23	↓	↓	↓	↓
D3	3C	↓	37, 16	↓	14	↓	↓	↓	↓
D4	4C	Output voltage, high level	17, 38	Open	34	0.0	-0.5	V	8

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TABLE VII. Output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	1C	Output voltage, low level	Open	26	3	-3.0	-5.0	V	8
E2	1C			6	3				
E3	2C			27	23				
E4	2C			7	23				
E5	3C			37	14				
E6	3C			16	14				
E7	4C			17	34				
E8	4C	Output voltage, low level	Open	38	34	-3.0	-5.0	V	8

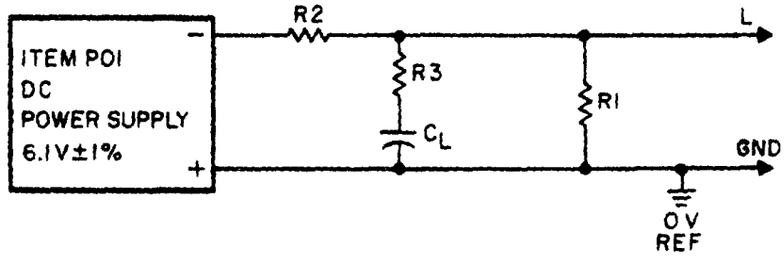
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. After connecting points A, B, and C and prior to making each measurement, apply point D to point C for a minimum of 1 second, then remove point D from point C.

FIGURE 8. Output voltage (high-level and low-level) test circuit for tables VI and VII

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NOTES:
1. R1 = 2.05 kΩ ±1%, 1/4 W.
2. R2 = 3.16 kΩ ±1%, 1/4 W.
3. R3 = 51Ω ±1%, 1/4 W.
4. C_L = 0.1 µF ±10%.
5. R3 and C_L are not required for dc electrical tests D1-D4 and E1-E8.
6. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, the 6 1 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

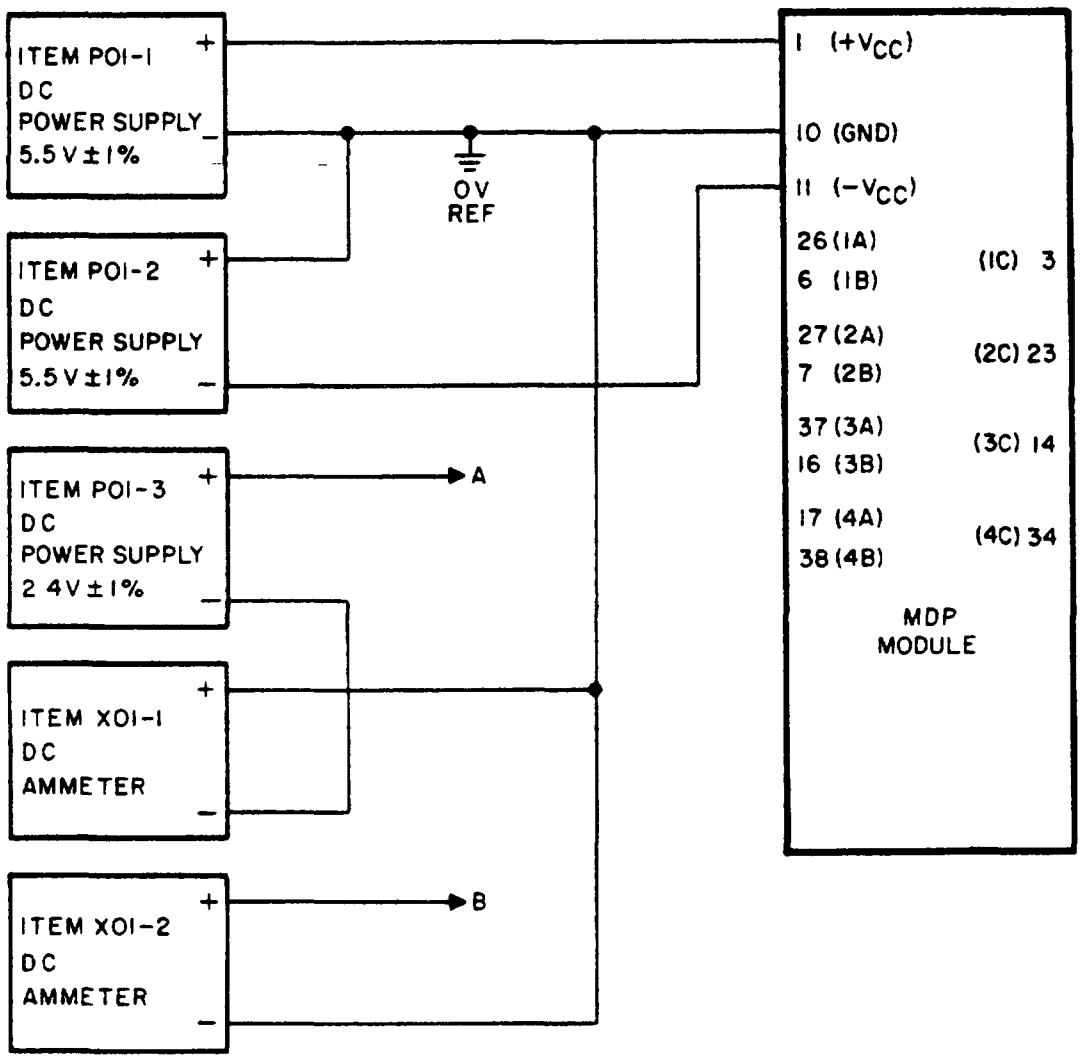
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	1A	Input current, low level	Open	26	0.4	1.73	mA	10
F2	1B	↓		6	↓	↓	↓	↓
F3	2A			27				
F4	2B			7				
F5	3A			37				
F6	3B			16				
F7	4A			17				
F8	4B	Input current, low level	Open	38	0.4	1.73	mA	10

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	1A Input current, high level	26	Open	0	30	µA	10
G2	1B	6					
G3	2A	27					
G4	2B	7					
G5	3A	37					
G6	3B	16					
G7	4A	17					
G8	4B Input current, high level	38	Open	0	30	µA	10

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. Since items XOI-1 and XOI-2 are not used simultaneously, only one item XOI is required. Use item XOI-2 for table VIII; use item XOI-1 for table IX.
2. With point A disconnected, item XOI-1 should indicate zero current on the 1 µA range.

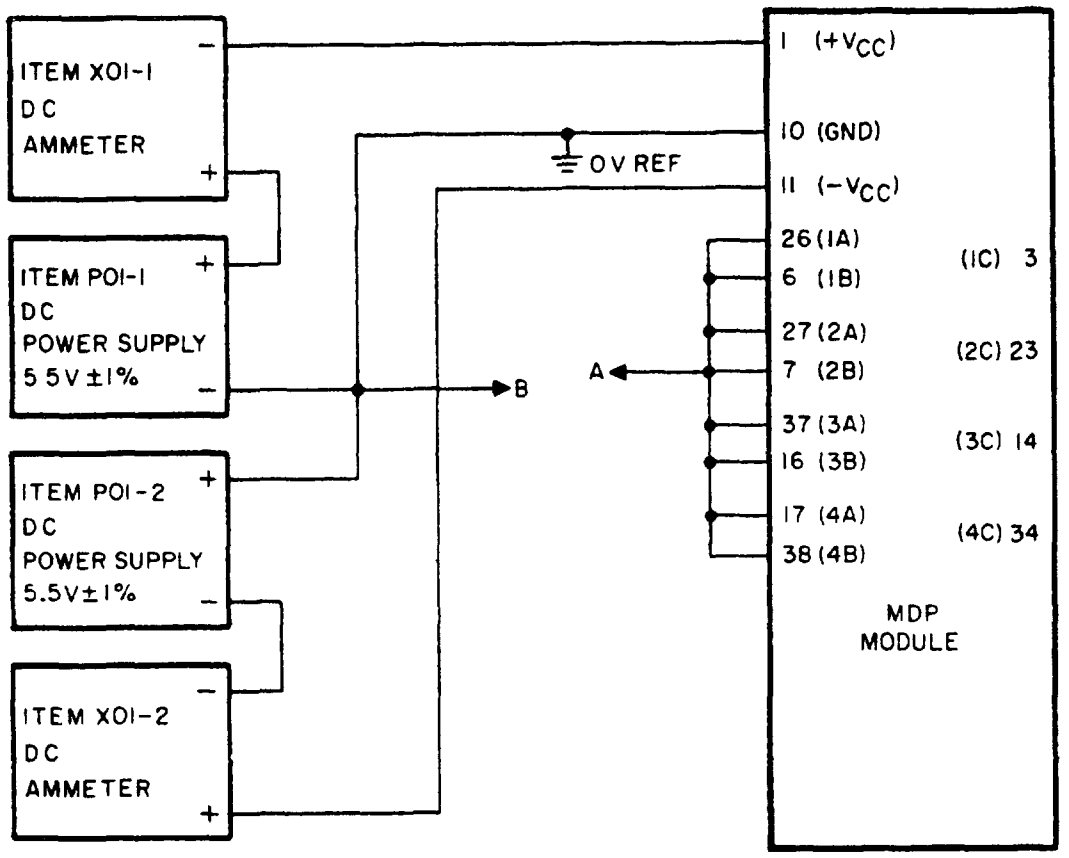
FIGURE 10. Input current (low-level and high-level) test circuit for tables VIII and IX.

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TABLE X. Power supply current (high-level and low-level at $+V_{CC}$ and $-V_{CC}$) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Power supply current, high level, $+V_{CC}$	Open	Open	1	92	mA	11
H2	Power supply current, high level, $-V_{CC}$	Open	Open		28		
H3	Power supply current, low level, $+V_{CC}$	B	A		80		
H4	Power supply current, low level, $-V_{CC}$	B	A	1	48	mA	11

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CAUTION.

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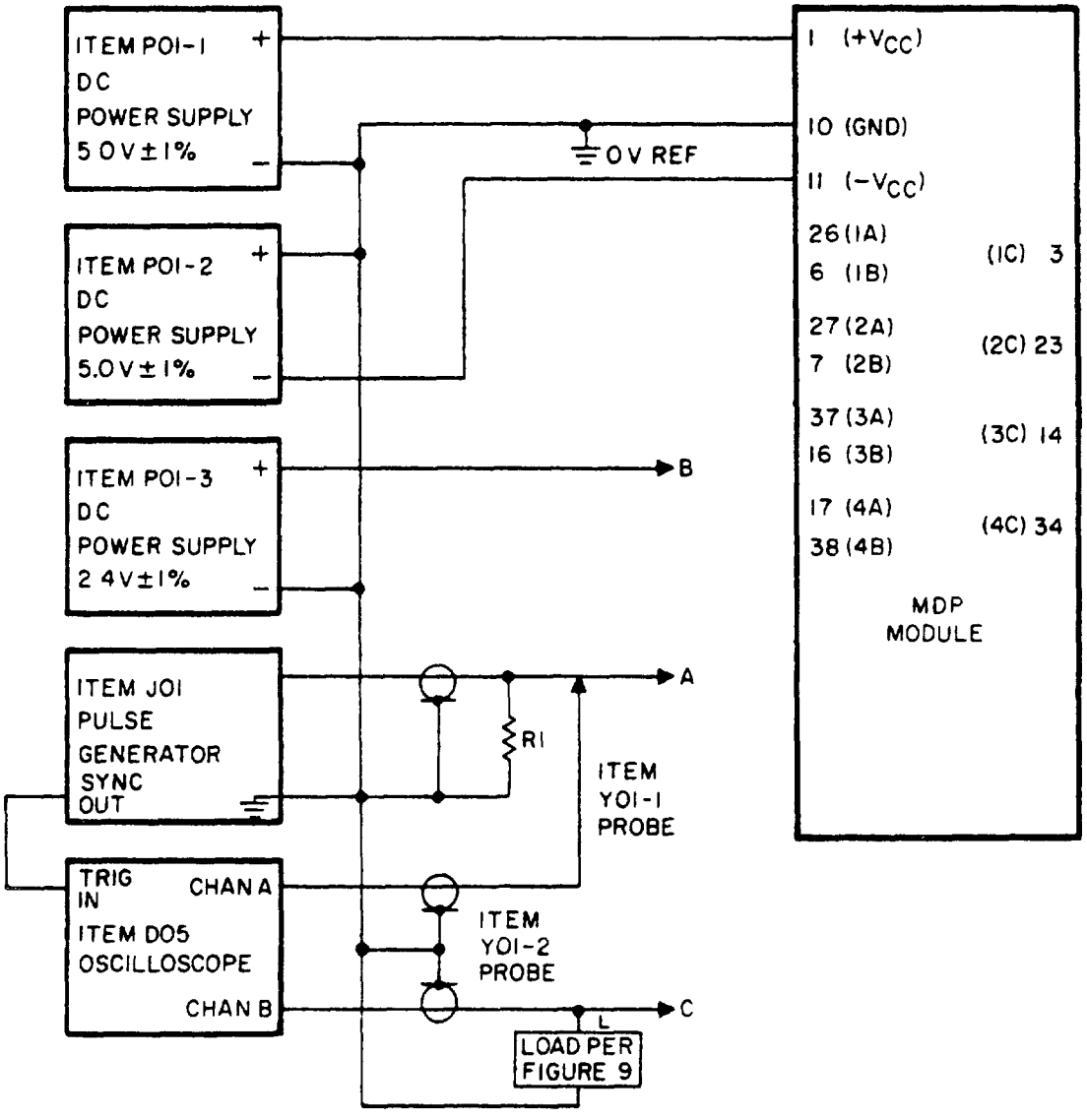
FIGURE 11. Power supply current (high-level and low-level at $+V_{CC}$ and $-V_{CC}$) test circuit for table X.

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TABLE XI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Transition time (t_{TLH})	6	26	3	4	40	ns	12, 13
J2	↓	7	27	23	↓	↓	↓	↓
J3	↓	16	37	14	↓	↓	↓	↓
J4	Transition time (t_{TLH})	38	17	34	↓	↓	↓	↓
K1	Transition time (t_{THL})	6	26	3	↓	↓	↓	↓
K2	↓	7	27	23	↓	↓	↓	↓
K3	↓	16	37	14	↓	↓	↓	↓
K4	Transition time (t_{THL})	38	17	34	4	40	↓	↓
L1	Propagation delay time (t_{PHL})	6	26	3	25	150	↓	↓
L2	↓	7	27	23	↓	↓	↓	↓
L3	↓	16	37	14	↓	↓	↓	↓
L4	Propagation delay time (t_{PHL})	38	17	34	↓	↓	↓	↓
M1	Propagation delay time (t_{PLH})	6	26	3	↓	↓	↓	↓
M2	↓	7	27	23	↓	↓	↓	↓
M3	↓	16	37	14	↓	↓	↓	↓
M4	Propagation delay time (t_{PLH})	38	17	34	25	150	ns	12, 13

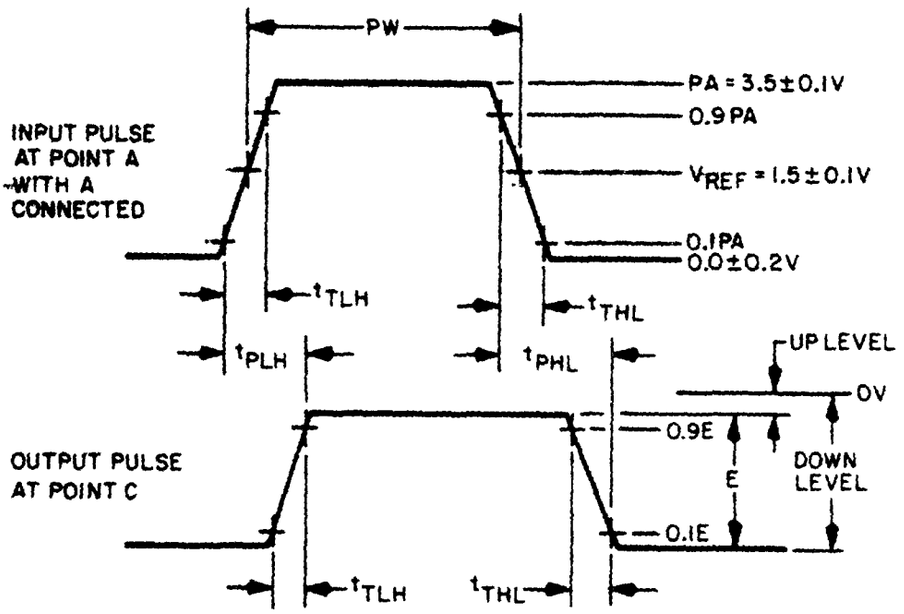
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NOTE. R1 = 51Ω ±5%, 1 W.

FIGURE 12 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XI

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NOTES:
1. Input pulse parameters:
   PW = 1 µs ±10%.
   PRR = 500 kHz ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making measurements, verify the output levels to be not more negative than -0.5 V for the "up" level and to be not less negative than -3.0 V for the "down" level.

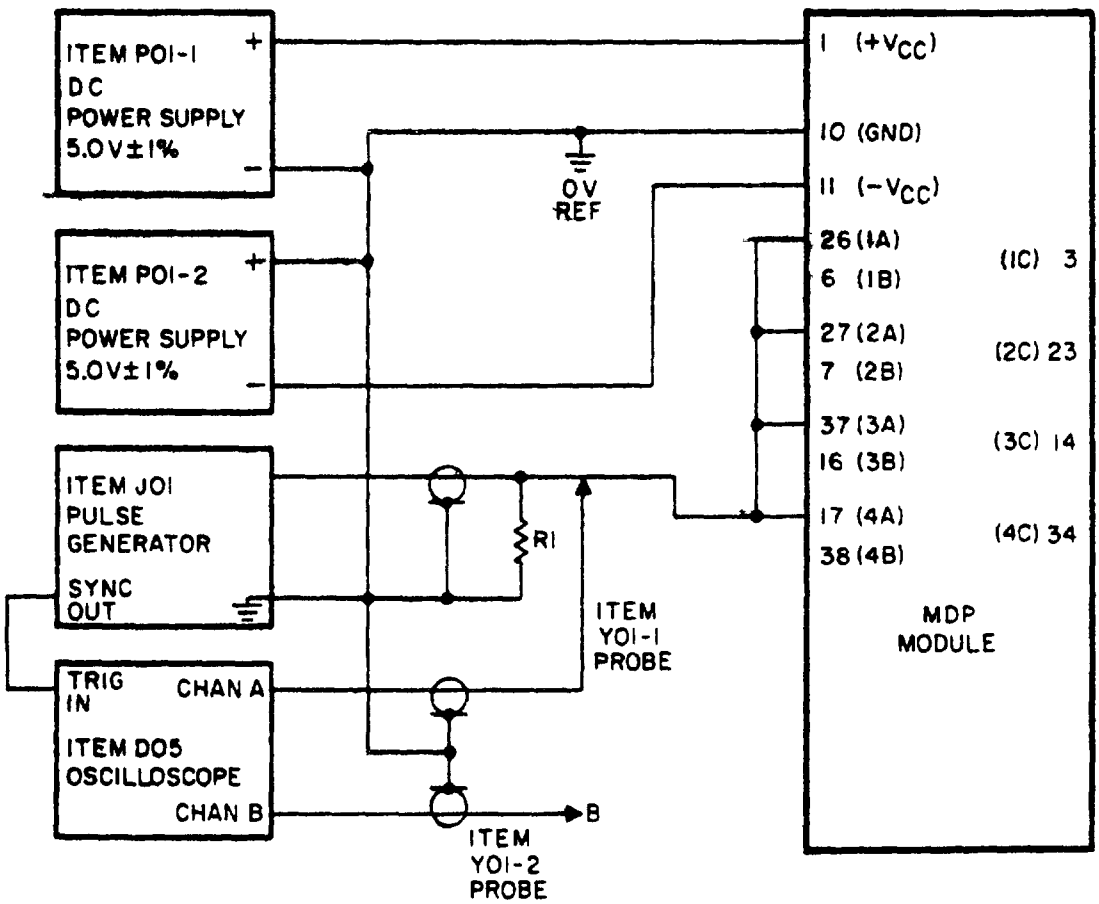
FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XI.

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TABLE XII. Power supply transient amplitude ($+V_{CC}$ and $-V_{CC}$) test.

Test condition	Parameter	Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
		R to pin	Min	Max		
N1	Power supply transient amplitude, $+V_{CC}$	1	0	600	mV	14, 15
N2	Power supply transient amplitude, $-V_{CC}$	11	0	500	mV	14, 15

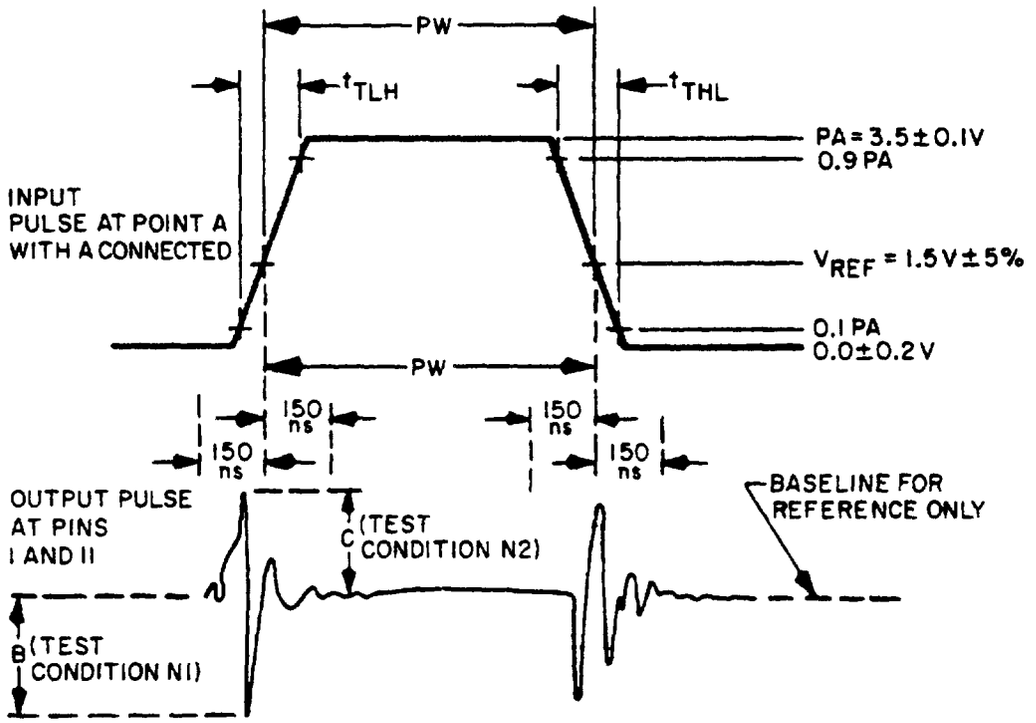
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 14. Power supply transient amplitude ($+V_{CC}$ and $-V_{CC}$) test circuit for table XII.

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NOTES
1. Input pulse parameters
   PW = 1 µs ±10%
   PRR = 500 kHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Amplitude B is measured from the baseline to the most negative excursion of the pulse, which may occur in time coincidence ±150 ns with the 1.5 Vdc point of either the leading edge or the trailing edge of the input pulse.
3. Amplitude C is measured from the baseline to the most positive excursion of the pulse, which may occur in time coincidence ±150 ns with the 1.5 Vdc point of either the leading edge or the trailing edge of the input pulse.

FIGURE 15. Power supply transient amplitude ($+V_{CC}$ and $-V_{CC}$) test circuit waveforms for table XII.

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TABLE XIII. Output crosstalk (low-level) test.

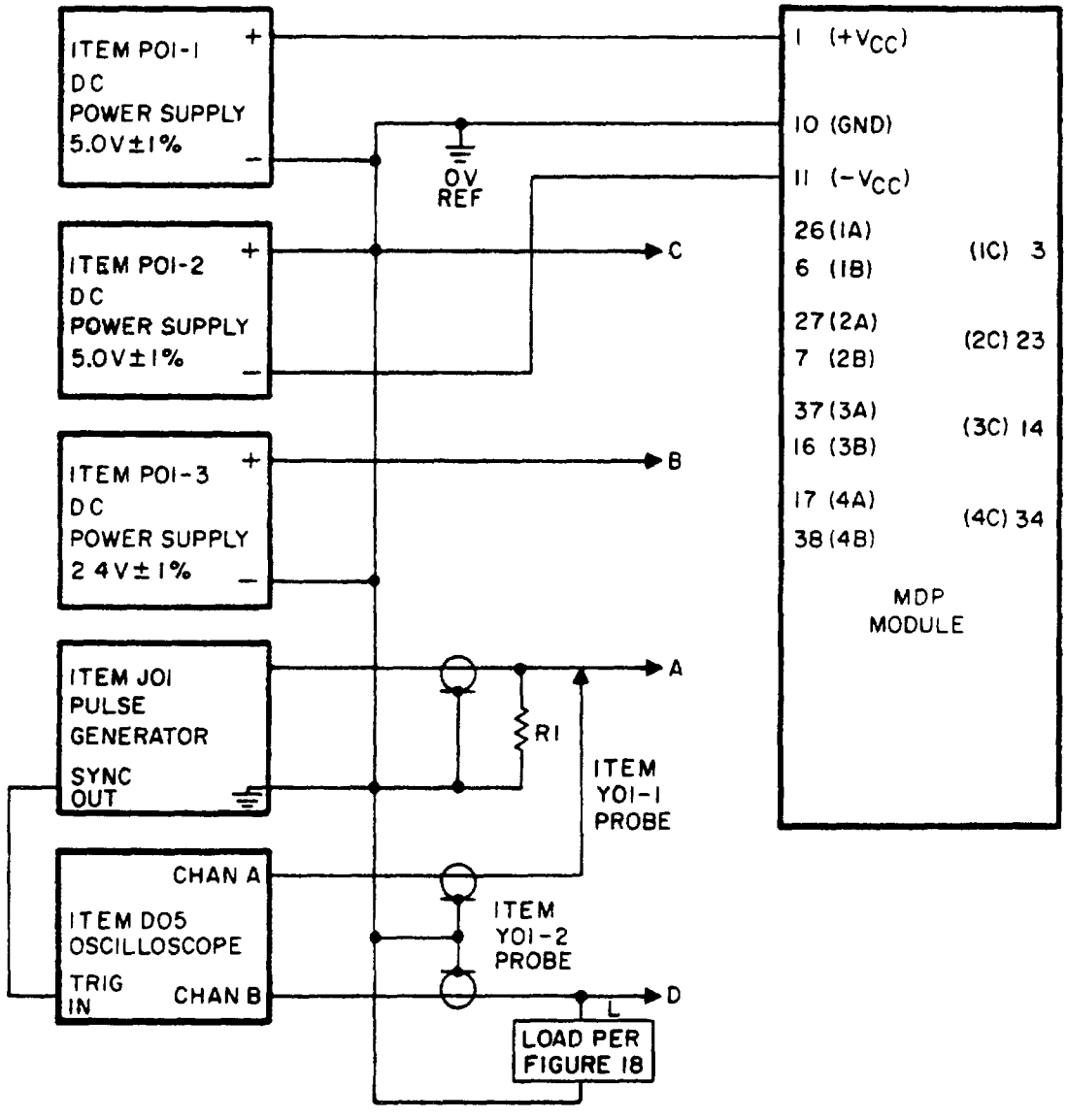
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Output crosstalk, low level	7,16 17	27,37 38	6,26	3	0	400	mV	16, 17
P2		6,16 17	26,37 38	7,27	23				
P3		6,7, 17	26,27 38	16,37	14				
P4	Output crosstalk, low level	6,7, 16	26,27, 37	17,38	34	0	400	mV	16, 17

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TABLE XIV. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Output crosstalk, high level	7,16 17	6,26, 27,37, 38	Open	3	0	400	mV	16, 17
R2		6,16, 17	7,27, 26,37, 38		23				
R3		6,7, 17	16,37, 26,27, 38		14				
R4	Output crosstalk, high level	6,7, 16	17,38, 26,27 37	Open	34	0	400	mV	16, 17

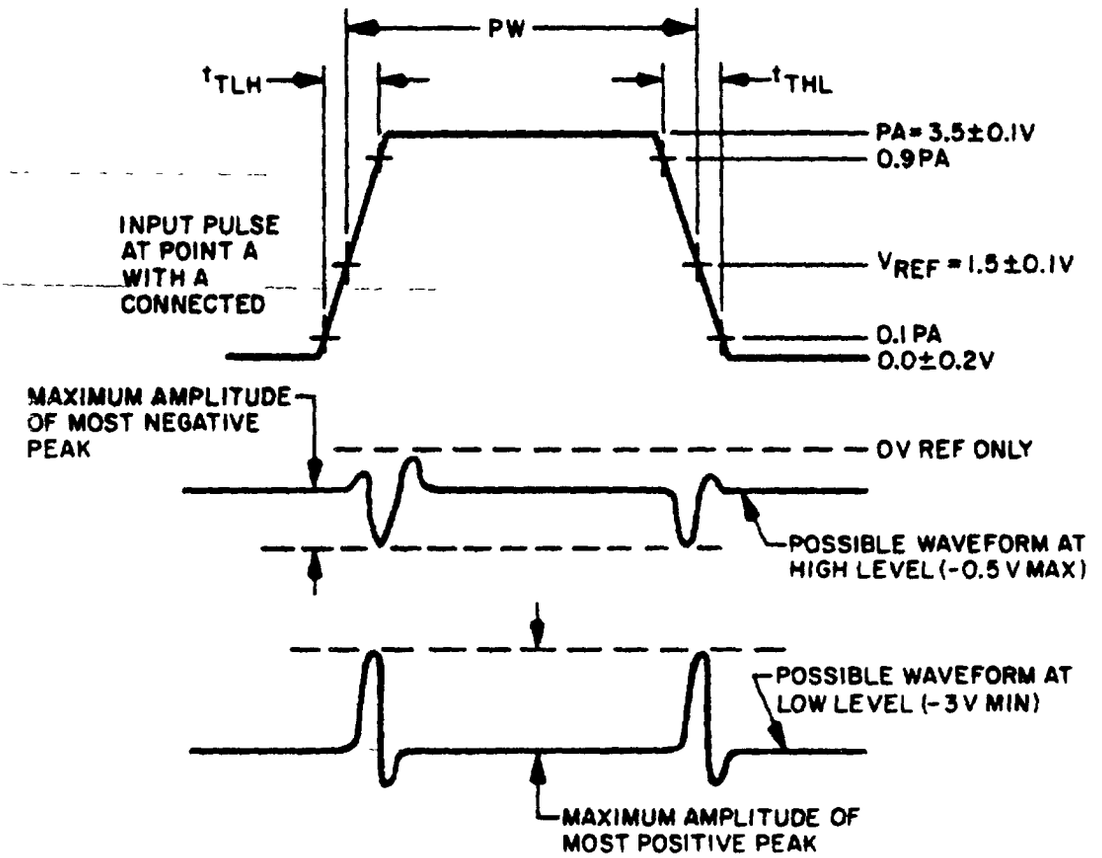
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Only the output under test shall be loaded.

FIGURE 16 Output crosstalk (low-level and high-level) test circuit for tables XIII and XIV.

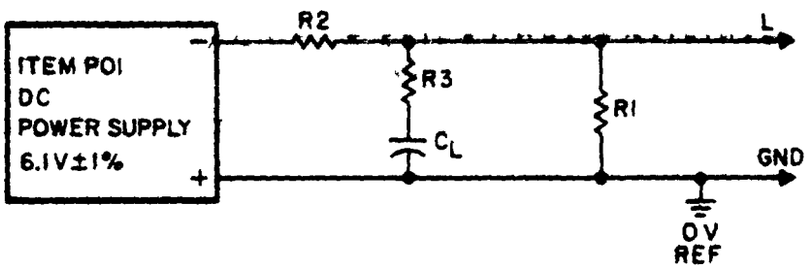
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NOTE: Input pulse parameters:
PW = 1 µs ±10%.
PRR = 500 kHz ±10%.
$t_{TLH} = t_{THL}$ = 25 ns ±10%.

FIGURE 17. Output crosstalk (low-level and high-level) test circuit waveforms for tables XIII and XIV.

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NOTES:
1. R1 = 3.01 kΩ ±1%, 1/4 W.
2. R2 = 5.11 kΩ ±1%, 1/4 W.
3. R3 = 147Ω ±1%, 1/4 W.
4. C_L = 0.01 µF ±10% (including probe and parasitic capacitance).
5. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2-mm).
6. During output switching, the 6.1 V power supply shall have a maximum overshoot of 500 mV.

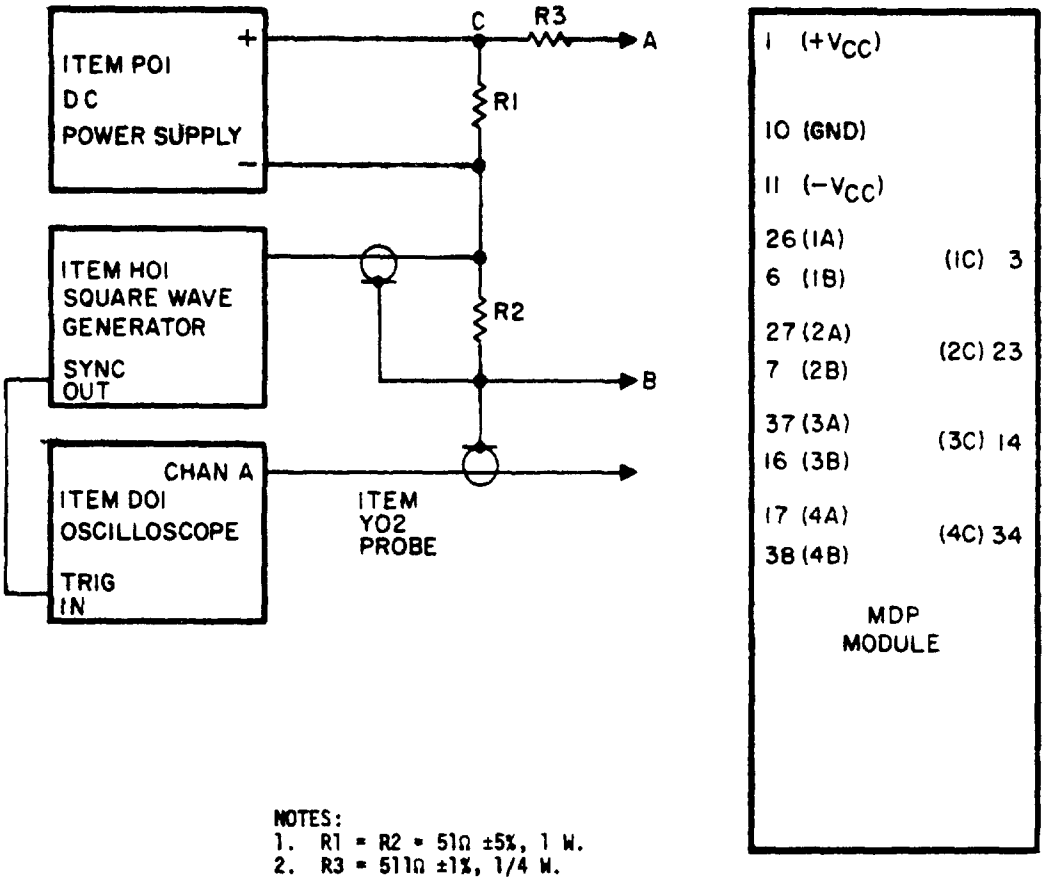
FIGURE 18. Logic load circuit for output crosstalk (low-level and high-level) for tables XIII and XIV.

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TABLE XV. Power supply transient rise time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	Power supply transient rise time	1	10, 11	8	12	ms	19, 20
S2	Power supply transient rise time	1,10	11	8	12	ms	19, 20

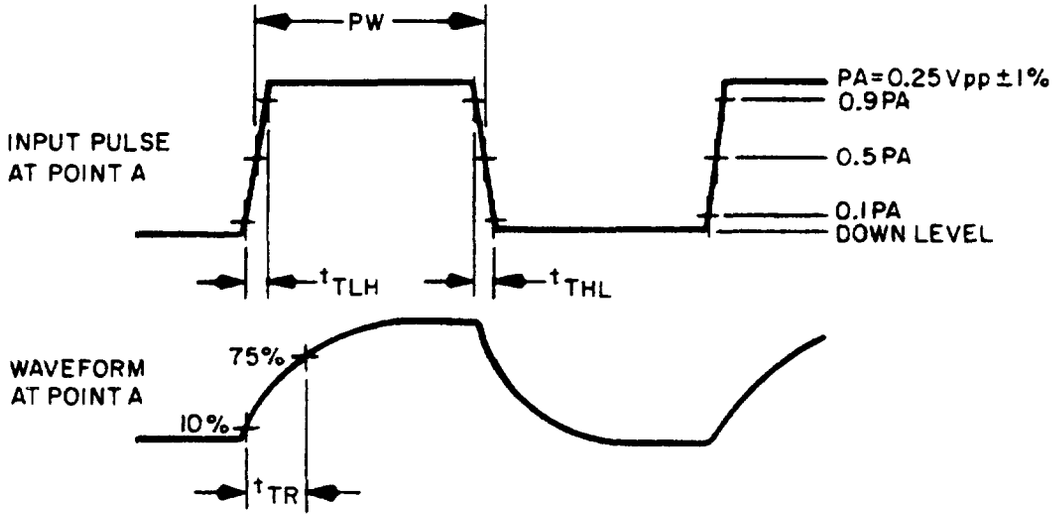
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 19. Power supply transient rise time test circuit for table XV.

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NOTES
1. Input pulse parameters:
   PW = 50 ms ±10%.
   PRR = 10 Hz (reference).
   $t_{TLH} = t_{THL} < 100$ ns (reference).
2. With point A open, adjust input as shown at point C
3. With point A open, adjust item P01 such that the down level of the input is at 0.00 ±0.01 V.
4. Connect points A and B to pins specified in table XV Recheck down level and pulse parameters at point C.
5. Measure transient rise time (t_{TR}) at point A; the values measured shall comply with test conditions S1 and S2.

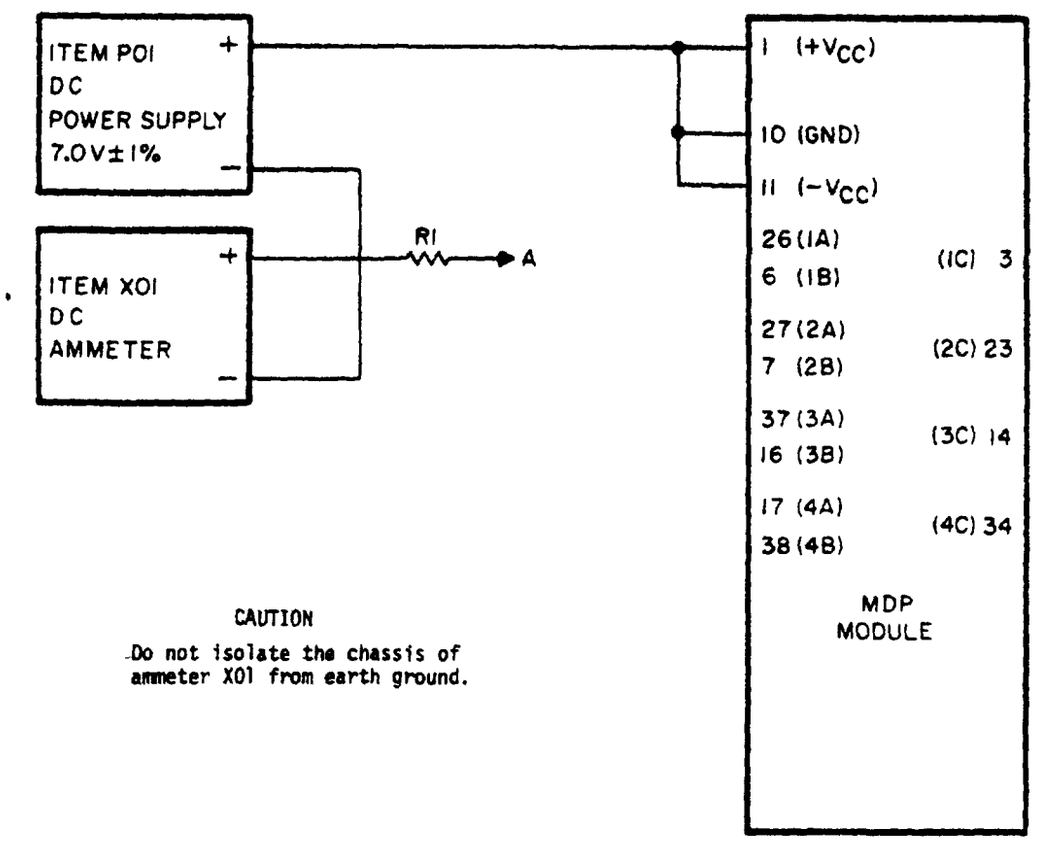
FIGURE 20. Power supply transient rise time test circuit waveforms for table XV.

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TABLE XVI. Output impedance check.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
T1	Output impedance check	3	0	70	µA	21
T2	↓	23	↓	↓	↓	↓
T3	↓	14	↓	↓	↓	↓
T4	Output impedance check	34	0	70	µA	21

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NOTE: R1 = 1.87 kΩ ±1%, 1/4 W.

FIGURE 21. Output impedance check test circuit for table XVI.

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